

The Gesture Lives On: A VR-Driven Puppet Performance in Immersive Space

Yi Jen Lin
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan
oxygendancevina@gmail.com

Wei-Chen Yen
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan
wilsonnosliw885@gmail.com

Chun-Cheng Hsu*
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan
cchsu@nycu.edu.tw

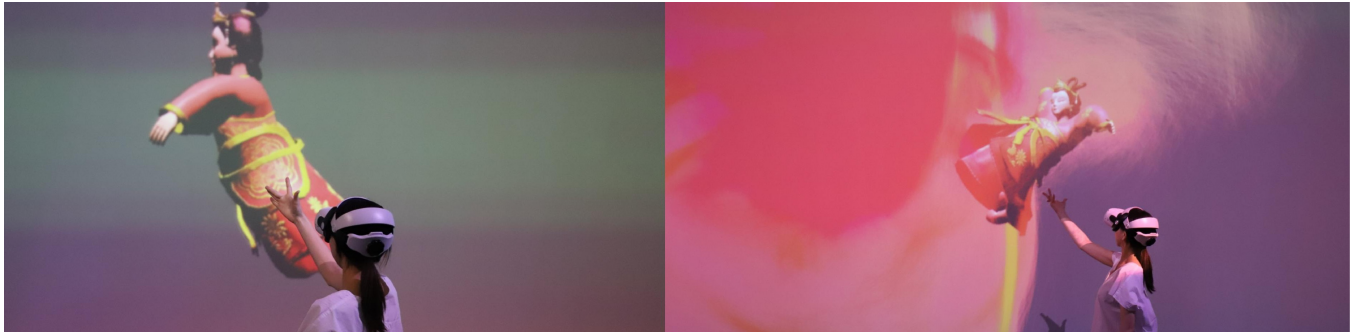


Figure 1: A performer controls a virtual puppet in real time through VR gesture tracking, generating dynamic visual narratives that reinterpret traditional Taiwanese glove puppetry within an immersive environment.

Abstract

The Gesture Lives On is a real-time VR performance system that reimagines traditional Taiwanese glove puppetry through immersive, interactive means. Rooted in precise gestural movement, this art form faces decline amid shifting audience engagement. Using VR-based gesture recognition, a performer co-creates a digital duet with a virtual puppet. This hybrid space transforms tradition into contemporary expression, offering audiences a new way to experience puppetry within a shared virtual-physical environment.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; • **Applied computing** → **Performing arts**.

Keywords

virtual reality, hand-gesture, dance, immersive performance, puppetry show

ACM Reference Format:

Yi Jen Lin, Wei-Chen Yen, and Chun-Cheng Hsu. 2025. The Gesture Lives On: A VR-Driven Puppet Performance in Immersive Space. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters*

*Corresponding Author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH Posters '25, Vancouver, BC, Canada
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1549-5/25/08
<https://doi.org/10.1145/3721250.3742981>

(SIGGRAPH Posters '25), August 10–14, 2025, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3721250.3742981>

1 INTRODUCTION

Taiwanese glove puppetry (Budaixi) is a centuries-old art form rooted in symbolic gestures and stylized movement. As a form of embodied storytelling, it conveys rich cultural meaning but now faces declining popularity amid shifting media and audience engagement patterns. With the development of motion-tracking and virtual reality technologies, recent years have seen increasing efforts to reimagine glove puppetry within digital contexts. A Leap Motion-based system focusing on gesture replication was developed to explore the possibilities of digital puppetry [Lin et al. 2018]. A cloud-based multi-user puppetry platform was later proposed to support remote collaboration [Way et al. 2019]. More recently, Theater of Future Puppetry [Yen et al. 2024] demonstrated the potential of integrating high-precision gesture control with immersive performance environments. Within this context, The Gesture Lives On presents a real-time VR performance system that reinterprets traditional puppet gestures through embodied action. Using gesture tracking, the performer interacts live with a responsive virtual puppet in a human-puppet duet. Immersive spatial projection extends these movements into a hybrid stage, proposing a new digital performance model that merges cultural heritage, choreography, and human-computer collaboration.

2 DESIGN AND IMPLEMENTATION

In traditional Taiwanese glove puppetry, classic character types include Sheng (male), Dan (female), Jing (painted face), Chou (comic), and Za (miscellaneous), each embodying distinct personalities and

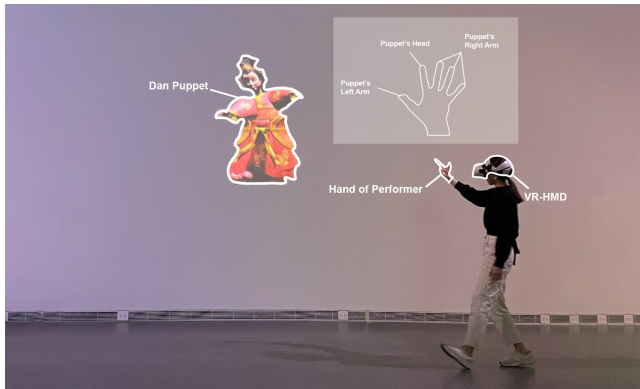


Figure 2: The performer controls the virtual Dan puppet's head and arms via traditional gesture mappings and VR hand tracking.

performance styles. This system centers on the Dan character, known for graceful and emotionally expressive female roles, with gestures such as “wiping tears,” “walking gently,” and “dancing elegantly.” For gesture control, the system is based on Dan character hand movements and utilizes the hand-tracking capabilities of the Oculus Quest 2 [Han et al. 2020]. Following traditional glove puppetry manipulation techniques, the system is designed so that the performer’s index finger controls the puppet’s head, the thumb controls the puppet’s left hand, and the middle finger controls the puppet’s right hand (Figure 2). Skeletal hand data is captured in real time through the headset’s built-in cameras without requiring external sensors. Motion feature extraction and precise mapping are conducted within the Unity platform, allowing the performer to control the virtual Dan puppet through natural gestures, achieving high synchronization between the performer’s movements and the puppet’s behavior. To dynamically visualize the performer’s hand movement trajectories, the system implements a real-time generation mechanism for flowing red particle trails. When the performer manipulates the virtual puppet, dynamic red trails are generated, extending gestures into visible spatial forms (Figure 3). These particle trails evolve in real time according to the speed, direction, and rhythm of the performer’s movements and are simultaneously projected into the immersive environment. Through this mechanism, every subtle gesture variation is captured, amplified, and presented as continuous, extensible trajectories across both the virtual and physical spaces, enhancing the temporal, spatial, and dynamic dimensions of the performance. The system emphasizes the performer’s active control within VR. Every movement is sensed in real time and translated into the puppet’s behavior and visual output. Virtual reality here serves not merely as a perceptual medium but as a performative tool, creating a digital puppetry model merging virtual and physical realms.

3 PERFORMANCE

The *Gesture Lives On* is a real-time performance that combines VR and live dance. Gesture recognition, puppet control, and particle effects are integrated into the choreography, with the system



Figure 3: Real-time red particle trails visualize the dynamics and spatial trajectories of the performer’s gestures behind the virtual puppet.

functioning as an extension of the performer’s body to form a feedback loop between action and perception. This responsive design fosters immersive engagement, enabling the performer to interact continuously with the system through real-time sensory feedback. During the performance, the dancer moves alone in a dark immersive space illuminated by flowing red light. As she raises her arm, the virtual Dan character appears and responds in real time—each gesture, turn, and step mirrored in perfect synchrony, forming a live duet across virtual and physical realms. As the performance progresses, trails of red particles emerge from the puppet, shifting with the dancer’s speed, direction, and rhythm. From subtle finger bends to sweeping turns, each movement leaves behind visual traces, painting rhythm and emotion into the immersive space. The audience is situated within a four-sided immersive projection environment, free to observe the dancer–puppet interaction from any angle. Real-time red particle trails visualize the rhythm and detail of each movement, transforming gestures into spatial narratives where motion and environment merge, and culture and technology are reimaged.

4 FUTURE WORK

The current system supports real-time co-performance between a single dancer and a virtual Dan character. Future developments will explore multi-performer configurations and audience participation, enabling collaborative storytelling and dynamic stage interaction through mobile or wearable devices. These extensions aim to transform the system from a standalone artwork into an open platform bridging traditional glove puppetry with contemporary performance technologies.

References

- Shangchen Han, Beibei Liu, Randi Cabezas, Christopher D Twigg, Peizhao Zhang, Jeff Petkau, Tsz-Ho Yu, Chun-Jung Tai, Muzaffer Akbay, Zheng Wang, et al. 2020. MEgATrack: monochrome egocentric articulated hand-tracking for virtual reality. *ACM Transactions on Graphics (ToG)* 39, 4 (2020), 87–1.
- Chi-Yen Lin, Zhang-Hao Yang, Heng-Wei Zhou, Tsai-Ni Yang, Hong-Nien Chen, and Timothy K Shih. 2018. Combining leap motion with unity for virtual glove puppets. In *2018 IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR)*. IEEE, 251–255.
- Der-Lor Way, Weng-Kei Lau, and Tzu Ying Huang. 2019. Glove puppetry cloud theater through a virtual reality network. In *ACM SIGGRAPH 2019 Posters*. 1–2.
- Wei-Chen Yen, Jen-Kai Liu, Jia-Jiu Peng, Hsiao-Yu Luo, Ping-Hsuan Han, and Chun-Cheng Hsu. 2024. Theater of Future Puppetry: An Immersive Puppeteering Experience Based on Hands Tracking and Gestures Recognition. In *SIGGRAPH Asia 2024 XR*. 1–2.